RGB-DPL

Manual v1.1.0

Body-Lights boards for Astromech droids

Firmware v6.0 (ESP32-S3) / v5.2 (Nano)
Platform Arduino Nano (ATmega328)

Hardware & Firmware

The **RGB-DPL** board and its firmware are developed and maintained by **Printed-Droid** for the R2 Builders community. Uses the **FastLED** library for WS2811/WS2812 driving.

Doku v1.1.0 — 2026-10-15

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- You are solely responsible for all work on your system.
- You have basic soldering and electronics knowledge or seek qualified help.
- You verify voltage, polarity and wiring before every power-on.
- You use only matching components (5V supply, correct LED strips, etc.).

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Table of Contents

Disclaimer and Liability Notice 2

Table of Contents 3

RGB-DPL — User Manual 6

■ Quickstart in 5 minutes . . . 6
Table of Contents . . . 6
1. What is RGB-DPL . . . 7
Three controller variants . . . 7
What runs on the RGB-DPL board itself — what is external? . . . 8
Other variant: DPL Classic / DPL-VU . . . 9
2. First setup . . . 9
Setup steps . . . 9
3. Hardware connections & pin map . . . 9
3.1 System architecture — RGB-DPL as central control . . . 10
3.2 External LED modules . . . 10
3.3 PCB breakouts on the RGB-DPL . . . 13
3.4 Complete pin map . . . 13
3.5 Notes for own development . . . 15
4. Flashing the firmware . . . 16
Pro Mini variant (firmware v5.2) . . . 16
ESP32-S3 Mini variant (firmware v6.0) . . . 16
ESP32-WROOM variant (RGB-DPL32, firmware v6.1) . . . 17
After flashing . . . 17
5. Command syntax . . . 17
How to configure a profile — step by step . . . 18
6. LED strips A through G . . . 18
LED sections for the COLOR command . . . 19
7. Animation modes . . . 19
LDPL (Strip E, 43 LEDs) . . . 19
CBI / D-CBI (Strip F, 20 matrix LEDs + 3 status) . . . 20
CSL (Strip G, 36 LEDs — external module) . . . 20
TopBlocks (Strip D, 18 LEDs) . . . 21
Bargraph (within Strip C, 18 LEDs) . . . 21

VU meter (Strip B, 28 LEDs) . . . 21
Quick commands (single-character hotkeys) . . . 21
8. Color schemes . . . 22
9. Personality modes . . . 22
10. Profiles (SAVE / LOAD / DEFAULT) . . . 23
Commands . . . 23
Default values . . . 23
11. Configuring the voltage monitor . . . 24
Activate . . . 24
Set voltage thresholds . . . 24
Voltage divider values (hardware adaptation) . . . 24
Display . . . 25
12. Door sensors (LeftDoor / RightDoor) . . . 25
How to activate the switches . . . 25
Behavior table . . . 26
13. System commands . . . 26
14. Local web UI & access point . . . 26
Connecting . . . 26
Automatic shutdown (save power/RF) . . . 27
What the web UI controls . . . 27
Web UI/Wi-Fi and voltage measurement . . . 27
15. Troubleshooting . . . 27
Command not accepted (no "OK") . . . 27
LEDs don't light up . . . 27
Wrong colors . . . 27
Voltage display shows wrong value . . . 28
EEPROM settings lost after update . . . 28
Watchdog reset (ESP32-S3/WROOM) . . . 28
"ESP-ROM:esp32s3" / no serial output (ESP32-S3) . . . 28
16. FAQ . . . 28
17. Cheat sheet (quick reference) . . . 29
General . . . 29
17.1 ESP32-S3 / WROOM (v6.1) — Full command reference . . . 29
17.2 Pro Mini (v5.2) — Differences from the ESP32 variant (S3/WROOM) . . . 31
Example setups (for both variants) . . . 32
18. Appendix: Legacy firmware v5.1 / v5.8 . . . 32

v5.1 (Pro Mini) — predecessor of v5.2 32
v5.8 (ESP32-S3) — predecessor of v6.0 32

RGB-DPL — User Manual

Firmware v6.1 (ESP32-S3 / ESP32-WROOM) · v5.2 (Nano / Pro Mini) · Doc 1.1.0 · 2026-10-15

This manual is for users of the Printed-Droid **RGB-DPL** board — the fully RGB-capable Data Port Logics variant for R2-D2 and other Astromech droids. It covers setup, hardware connections, operation, all commands, all animations and troubleshooting.

■ Quickstart in 5 minutes

If you've just unpacked the board and want to see it running:

1. **5V** to the `5V IN` screw terminal — a USB cable from your PC is enough
2. **USB cable** to the on-board Pro Mini, ESP32-S3 or ESP32-WROOM (depending on variant)
3. **Bridge the door switch headers** — solder a short wire jumper between the two pins of both `Left Door` AND `Right Door` headers (otherwise some LED sections stay dark)
4. **Flash the firmware** with the Arduino IDE — see *Section 4*
5. **Open a serial terminal** (Pro Mini: 57600 baud, ESP32-S3/WROOM: 9600 baud, line ending: Newline) and send e.g. `BRIGHTNESS 50` — or, on ESP32-S3/WROOM, use the local web UI instead (see *Section 14*)

Working? Great. For commands, color schemes, personalities and external modules, read on. Not working? → *Section 14 Troubleshooting*, check the door switch jumpers first.

Table of Contents

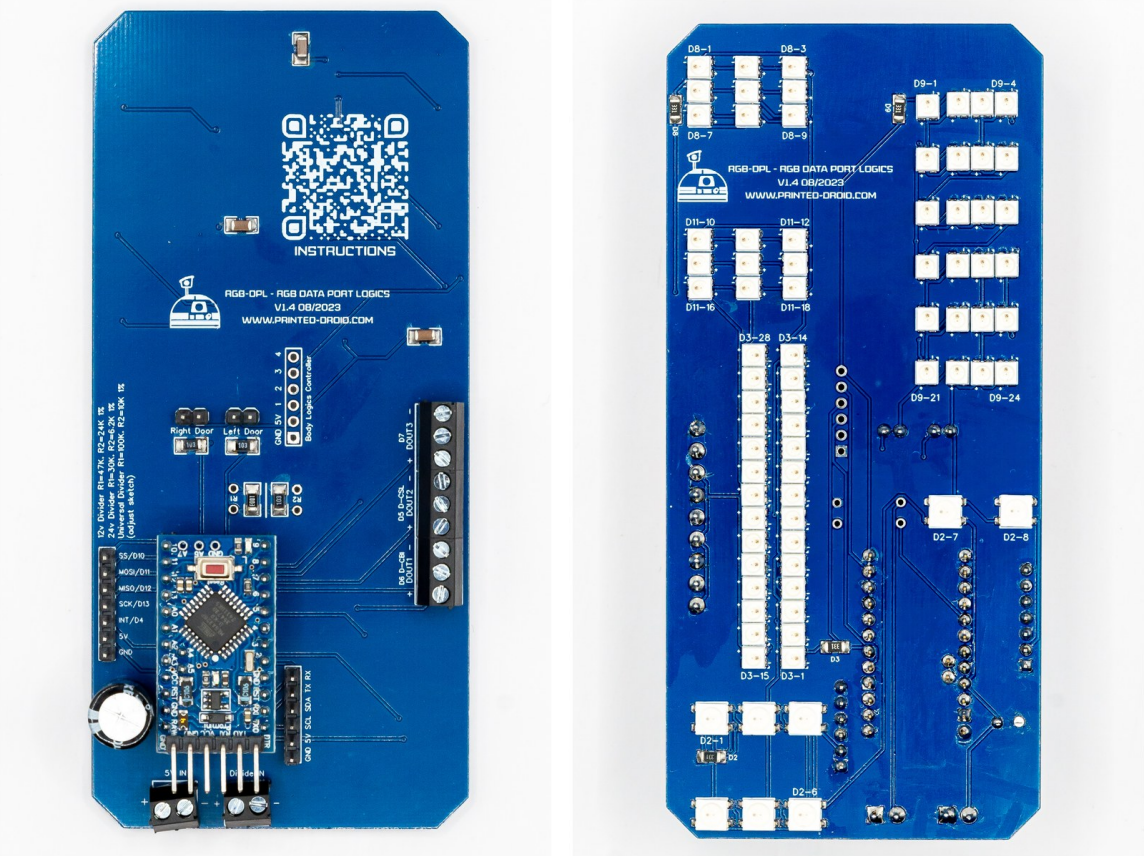
1. *What is RGB-DPL*
2. *First setup*
3. *Hardware connections & pin map*
4. *Flashing the firmware*
5. *Command syntax*
6. *LED strips A through G*
7. *Animation modes*
8. *Color schemes*
9. *Personality modes*
10. *Profiles (SAVE / LOAD / DEFAULT)*
11. *Configuring the voltage monitor*
12. *Door sensors (LeftDoor / RightDoor)*
13. *System commands*
14. *Local web UI & access point*
15. *Troubleshooting*
16. *FAQ*
17. *Cheat sheet (quick reference)*
18. *Appendix: Legacy firmware v5.1 / v5.8*

1. What is RGB-DPL

The **RGB-DPL** is an LED controller board for Astromech droids with **fully addressable WS2811/WS2812 LEDs** on all body panels:

- **DPL** — Data Port Lights
- **CBI** — Charge Bay Indicator
- **LDPL** — Large DataPanel Logics
- **CSL** — Coin Slot Logics

It replaces older MAX7219-based designs and gives you full color control — every LED can show any color.



RGB-DPL — back side (left) with connectors, Pro Mini socket and power terminal. Front side (right) with all on-board RGB LEDs (TopBlocks, bargraph, VU meter, bottom white, red large).

Three controller variants

Depending on the RGB-DPL board / hardware revision, a different microcontroller is used. Pro Mini and ESP32-S3 Mini sit on the **same** RGB-DPL board (only the socket differs); the **ESP32-WROOM** firmware `RGB-DPL32` targets the **older** RGB-DPL hardware (up to HW REV 1.3).

Feature	Pro Mini (Nano)	ESP32-S3 Mini	ESP32-WROOM (RGB-DPL32)
Microcontroller	Arduino Pro Mini (ATmega328P)	LOLIN ESP32-S3 Mini	ESP32-WROOM (D1-Mini form / classic ESP32)

Feature	Pro Mini (Nano)	ESP32-S3 Mini	ESP32-WROOM (RGB-DPL32)
Current firmware	**v5.2**	**v6.1**	**v6.1**
Matching board	RGB-DPL with Pro Mini socket	RGB-DPL **from HW REV 1.4**	RGB-DPL **up to HW REV 1.3**
LED strips	7 (A-G), 180 LEDs	7 (A-G), 180 LEDs	6 (A-F), 144 LEDs (**no CSL**)
Baud rate	57600	9600	9600
WiFi module	no (ATmega328P)	yes	yes
Local web UI / AP	no (no WiFi)	**yes**	**yes**
I2C slave	no	`0x20`	`0x20` (handler currently a stub)
Watchdog	no	10 s	10 s
LDPL modes	4	16	16
TopBlocks modes	2 (RANDOM / CLASSIC)	5 (0-4)	5 (0-4)
Headroom	very tight (flash 30k/30k)	plenty	plenty

All variants support:

- The on-board LED strips (A-F) plus D-CBI and LDPL as external modules; **CSL (strip G) only on Pro Mini and ESP32-S3**, not the WROOM.
- 5 profiles (`SAVE 1-5`, `LOAD 1-5`, `DEFAULT 0-5`)
- 8 color schemes (CLASSIC, BLUE, PINK, GREEN, CYBERPUNK, FOREST, SUNSET, CUSTOM)
- 5 personality modes (NORMAL, HAPPY, GRUMPY, EXCITED, SLEEPY)
- 16 LDPL modes (ESP32-S3/WROOM from v6.1; Pro Mini still 4), 7 CBI modes, 8 CSL modes (only boards with CSL)
- Door switch inputs for automatic hatch detection
- Voltage monitor with configurable thresholds (`VM`, `VL`, `RR`)
- Sequence mode (auto-cycling animations)

What runs on the RGB-DPL board itself — what is external?

On the board itself (see image above):

- **DPL sections** (TopBlocks, bargraph, VU meter, blue column, bottom white, red large)

External modules (sold separately, plug into the DPL):

- **D-CBI** — digital Charge Bay Indicator (WS2812B, on D6 / IO9)
- **LDPL** — Large DataPanel Logics (WS2812B strip, on D7 / IO8)
- **CSL C3 Mini** — Coin Slot Logics (WS2812B, on D5 / IO13)

If an external module is not connected, the corresponding animation simply runs into the void in the firmware — no error, just nothing visible.

Other variant: DPL Classic / DPL-VU

If you're looking for a **classic variant** with MAX7219 LEDs (for DPL and CBI) mixed with WS2812B (for DCBI, LDPL, UAL), Printed-Droid also offers the **DPL Classic** and **DPL-VU**. They run the AstroCAN BodyLights firmware (by RealNobser) with slash-based commands (`/DP/BR/15`) — no profile system, but a simpler build with fixed-color analog LEDs.

Documentation & firmware: https://github.com/PrintedDroid/DPL-VU_and_DPL-Classic

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2. First setup

> The RGB-DPL has two door-switch inputs (`Left Door`, `Right Door`). As long as these are **not** connected, the firmware keeps certain LED sections **dark** — as if the hatches were closed. **That is intentional**, but accounts for 80% of all "doesn't work" reports.
> So when testing for the first time: either connect reed switches, **or** set a **wire jumper** between the two pins of each header. Details in *Section 12*.

Setup steps

1. **Power supply**
5V to the `5V IN` screw terminal. The board runs **exclusively on 5V** — no 12V on this input!
2. Optional **battery voltage** (12V / 24V) to the `Voltage IN` screw terminal. **This is ONLY a measurement input** for the battery indicator, not an additional power feed.
3. **Current draw:** With default brightness levels, a **standard USB cable directly from your PC** is sufficient as the only power source. Only at very bright setups (all 180 WS2812Bs at full white) does a separate 5V/4A supply make sense.
4. **Connect external LED modules** (all optional, depending on your build)
D-CBI → 3-pin connector `D6 D-CBI` on the DPL (5V / Signal / GND)
5. **LDPL** strip → 3-pin connector `D7 LDPL`
6. **CSL C3 Mini** → 3-pin connector `D5 D-CSL`
7. **Door switches** (see warning above — sections stay dark otherwise!)
Reed switch / micro-switch / hall sensor to the 2-pin headers `Right Door` and `Left Door`
8. Or for testing: wire jumper between the two pins of each header
9. Details and all variants in *Section 12*
10. **Plug a USB cable into the on-board microcontroller** (Pro Mini, ESP32-S3 or ESP32-WROOM, depending on your variant)
11. **First function check**
On first start a boot sequence plays (Pro Mini: 3× white flash; ESP32-S3/WROOM: rainbow wave + 3× white flash)
12. After that all strips run with their default animations
13. If nothing lights up: check the door switches, send `STATUS` in the terminal for diagnostics

---

3. Hardware connections & pin map

This section first explains the system architecture, then the actual connectors on the DPL board, and finally the complete pin assignments for both controller variants.

3.1 System architecture — RGB-DPL as central control

The RGB-DPL board is the **master board** with on-board microcontroller. All other LED modules are separate boards that connect to the DPL:

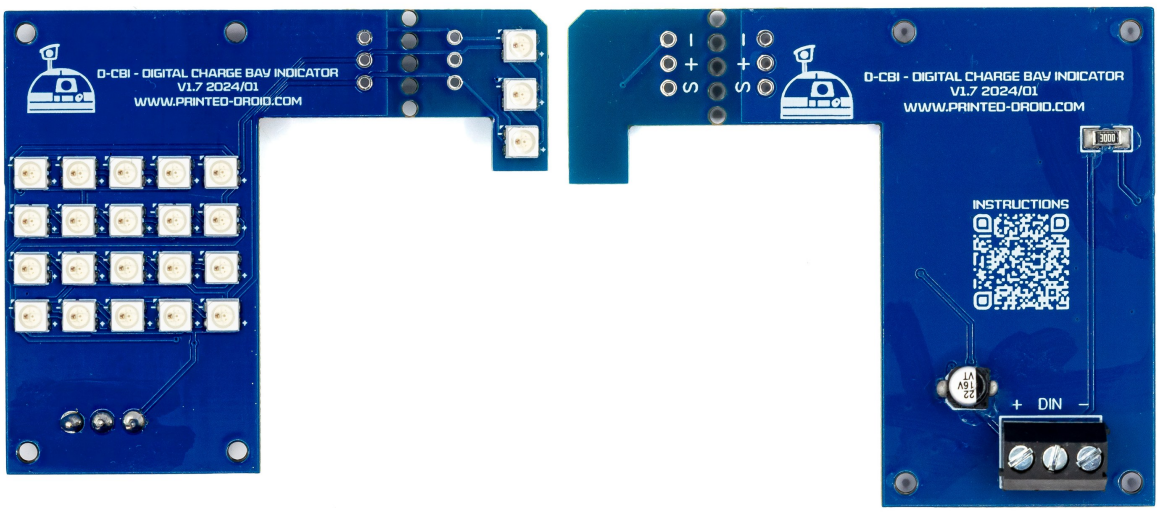
External module	Connector on the DPL	Command token	LED count
D-CBI (digital Charge Bay Indicator)	3-pin header `D6 D-CBI`	`CBIMODE`	23 (20 matrix + 3 status)
LDPL (Large DataPanel Logics)	3-pin header `D7 LDPL`	`LDPLMODE`	43
CSL C3 Mini (Coin Slot Logics)	3-pin header `D5 D-CSL`	`CSLMODE`	36

Important: The DPL's own LEDs (TopBlocks, bargraph, VU, etc.) are populated directly on the RGB-DPL board — these are **always** present and driven directly by the microcontroller.

3.2 External LED modules

The following modules are sold separately by Printed-Droid and connect to the RGB-DPL:

D-CBI (digital Charge Bay Indicator)



D-CBI — back side (left) with DIN screw terminal (`+ DIN −`) for the main connection to the RGB-DPL. Front side (right) with 23 WS2812B LEDs (20 matrix + 3 status LEDs).

- Full RGB color depth — every one of the 20 matrix LEDs can show any color
- Command: `CBIMODE 0-6` (Organic, ESB, Rainbow, Sparkle, Heart, Smiley, Matrix Rain)
- 3 status LEDs for the voltage indicator (red/yellow/green)
- Connects via the 3-pin terminal `D6 D-CBI` on the RGB-DPL

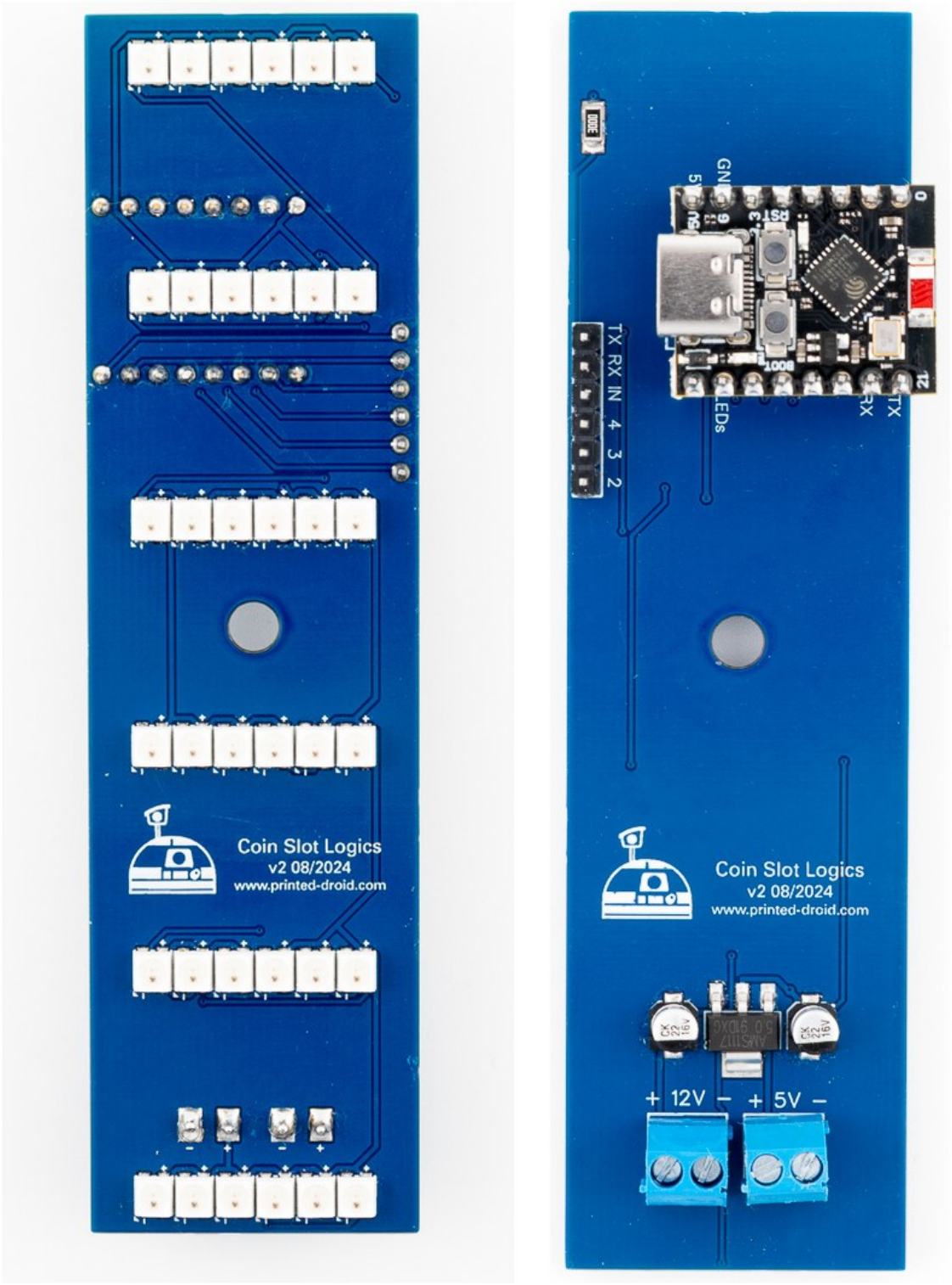
LDPL (Large DataPanel Logics)



LDPL — curved WS2812B strip with 43 LEDs for the belly hatch lighting. Center has a 3-pin screw terminal (`– S +`) for connection to D7 of the RGB-DPL.

- 43 WS2812B LEDs in a curved arrangement
- Command: `LDPLMODE 0-15` (Off, Dot bounce, Fade trail, Rainbow dot, KITT, Cylon, Beat bars, Breathing, Juggle, Sparkle, Comet, Alarm, Fire, Logic sweep, Rainbow bars, Theater chase — Pro Mini: `0-3`); set the base color with `LDPLCOLOR R G B`
- Full RGB color depth via color schemes or `COLOR 11 R G B` (custom)

CSL C3 Mini (Coin Slot Logics)



CSL C3 Mini — back side (left) with the on-board ESP32-C3 Mini controller. Front side (right) with 36 WS2812B LEDs.

The CSL C3 Mini has **its own ESP32-C3 Mini controller** and can therefore run standalone (without an RGB-DPL). It has two operating modes:

- **Standalone (default):** The on-board ESP32-C3 plays the CSL animations autonomously — no RGB-DPL needed. CSL configuration is done directly through its own interface.

• **Connected to the RGB-DPL:** If the CSL is built without the on-board controller (or that controller is disabled), the CSL can instead be wired into the 3-pin terminal `D5 D-CSL` on the RGB-DPL. The RGB-DPL firmware then drives it via the `CSLMODE 0-7`, `CSLSPEED 1-10` and `CSLBRIGHTNESS 1-100` commands.

> The CSL commands described in this manual apply to the **connected** mode (without its own CSL controller). In standalone mode, refer to the separate CSL C3 Mini documentation.

3.3 PCB breakouts on the RGB-DPL

The RGB-DPL exposes the following connectors:

Function	Pin (Pro Mini / ESP32-S3)	Form
Power In: **5V** + GND	—	Screw terminal
Voltage In (battery for measurement, NOT power!)	A0 / IO12	Screw terminal
3-pin LED-Out **D-CBI** (5V / Signal / GND)	D6 / IO9	Screw terminal
3-pin LED-Out **D-CSL** (5V / Signal / GND)	D5 / IO13	Screw terminal
3-pin LED-Out **LDPL** (5V / Signal / GND)	D7 / IO8	Screw terminal
Door switch Left (D2 + GND)	Pin 16 (A2) / IO10	2-pin header
Door switch Right (D4 + GND)	Pin 15 (A1) / IO11	2-pin header
I2C breakout (5V / SDA / SCL / GND)	A4/A5 (only ESP32 active)	Pin header
UART breakout (RX / TX)	D0/D1 / USB-CDC	Pin header

3.4 Complete pin map

This table is helpful for users who want to write their **own firmware** for the board.

Pin map Pro Mini variant (ATmega328P)

Nano/Pro Mini pin	Function	LED strip / connector	LEDs
D2	`DATA_PIN_A`	Strip A — Bottom White (6) + Red Large (2)	8
D3	`DATA_PIN_B`	Strip B — VU meter	28
D5	`DATA_PIN_G`	Strip G — CSL (external)	36
D6	`DATA_PIN_F`	Strip F — D-CBI matrix (20) + status (3)	23
D7	`DATA_PIN_E`	Strip E — LDPL (external)	43
D8	`DATA_PIN_D`	Strip D — TopBlocks (yellow 9 + green 9)	18
D9	`DATA_PIN_C`	Strip C — Blue column (6) + bargraph (18)	24
Pin 16 (A2)	`LEFT_DOOR_PIN`	Door switch left, INPUT_PULLUP	—
Pin 15 (A1)	`RIGHT_DOOR_PIN`	Door switch right, INPUT_PULLUP	—

Nano/Pro Mini pin	Function	LED strip / connector	LEDs
A0	`ANALOG_INPUT`	Voltage monitor (via voltage divider)	—
A4 (SDA), A5 (SCL)	(not active in v5.2)	I2C breakout	—
D0, D1	UART RX/TX	USB serial	—
D4, D10-D13	free	(for own code)	—

Total: 180 LEDs on 7 strips (A-G).

Pin map ESP32-S3 Mini variant (before HW REV 1.4)

ESP32-S3 GPIO	Function	LED strip / connector	LEDs
IO4	`DATA_PIN_A`	Strip A — Bottom White (6) + Red Large (2)	8
IO5	`DATA_PIN_B`	Strip B — VU meter	28
IO6	`DATA_PIN_C`	Strip C — Blue column (6) + bargraph (18)	24
IO7	`DATA_PIN_D`	Strip D — TopBlocks (yellow 9 + green 9)	18
IO8	`DATA_PIN_E`	Strip E — LDPL (external)	43
IO9	`DATA_PIN_F`	Strip F — D-CBI matrix (20) + status (3)	23
IO13	`DATA_PIN_G`	Strip G — CSL (external)	36
IO10	`LEFT_DOOR_PIN`	Door switch left, INPUT_PULLUP	—
IO11	`RIGHT_DOOR_PIN`	Door switch right, INPUT_PULLUP	—
IO12	`ANALOG_INPUT`	Voltage monitor (via voltage divider, ADC 12-bit)	—
Default SDA / SCL	I2C slave (address 0x20)	I2C breakout	—
USB-CDC	Serial	via on-board USB	—

> **Note on the voltage divider:** Default values are R1 = 100 kΩ and R2 = 10 kΩ (same divider as the DPL Classic). With the `RR <R1> <R2>` command you can adapt the software scaling to other hardware populations.

Pin map ESP32-S3 Mini variant (from HW REV 1.4)

From hardware revision 1.4 the DPL strips A-D are combined into **one chain on IO4**; LDPL, CBI and CSL stay separate:

ESP32-S3 GPIO	Function	LED strip / connector	LEDs
IO4	Chain A→D	A Bottom White (6) + Red Large (2) → B VU (28) → C Blue (6) + bargraph (18) → D TopBlocks (9+9)	78
IO8	`DATA_PIN_E`	Strip E — LDPL (external)	43

ESP32-S3 GPIO	Function	LED strip / connector	LEDs
IO9	`DATA_PIN_F`	Strip F — D-CBI matrix (20) + status (3)	23
IO13	`DATA_PIN_G`	Strip G — CSL (external)	36
IO10 / IO11	Door switch left / right, INPUT_PULLUP	—	—
IO12	`ANALOG_INPUT`	Voltage monitor	—

HW REV 1.4 extra: command `CSLDOOR 0-2` (CSL always on / only with the left / only with the right door switch). **Do not mix firmware between revisions** — the LED outputs are wired differently.

Pin map ESP32-WROOM variant (RGB-DPL32, up to HW REV 1.3)

The WROOM has **no CSL** (6 strips A-F, 144 LEDs). Fixed pin map of the RGB-DPL hardware up to HW REV 1.3:

ESP32-WROOM GPIO	Function	LED strip / connector	LEDs
25	`DATA_PIN_A`	Strip A — Bottom White (6) + Red Large (2)	8
27	`DATA_PIN_B`	Strip B — VU meter	28
32	`DATA_PIN_C`	Strip C — Blue column (6) + bargraph (18)	24
4	`DATA_PIN_D`	Strip D — TopBlocks (yellow 9 + green 9)	18
19	`DATA_PIN_E`	Strip E — LDPL (external)	43
13	`DATA_PIN_F`	Strip F — D-CBI matrix (20) + status (3)	23
16 / 17	Door switch left / right, INPUT_PULLUP	—	—
0	`ANALOG_INPUT`	Voltage monitor (ADC2)	—
Default SDA / SCL	I2C slave (`0x20`, handler stub)	I2C breakout	—

WROOM note: The voltage ADC is on GPIO0 (ADC2). ADC2 is dead while WiFi/AP is active — the voltage monitor is therefore **off by default** (`VM 1` to enable, after `AP OFF`/timeout).

3.5 Notes for own development

- **Pro Mini variant (ATmega328P):** Flash and SRAM are very tight — the firmware uses 30k/30k flash and 1825/2048 bytes RAM. Extensions need aggressive optimization. I2C is therefore disabled in v5.2.
- **ESP32-S3 variant:** Plenty of headroom (397k/1310k flash, 25k/327k RAM). I2C slave is active on address `0x20` and ISR-safe.
- **GND distribution:** All GND pins (screw terminals, headers, door switches, I2C, UART) are connected together — your own devices can use any GND pin.

4. Flashing the firmware

The RGB-DPL firmware is **flashed with the Arduino IDE**. Which variant you use depends on the microcontroller installed on the board.

Pro Mini variant (firmware v5.2)

Arduino IDE configuration:

- Board: **Arduino Nano**
- Processor: **ATmega328P** (or for some clones: ATmega328P (Old Bootloader))
- Port: pick the correct COM port
- Library: install **FastLED** (3.5.0 or newer)

Upload:

1. Connect Pro Mini to PC via USB-FTDI adapter (or built-in USB if present)
2. Open the sketch `RGB_DPL-Nano_ProMini_v5.2.ino`
3. Click **Upload**

Serial terminal after flashing:

- Baud rate: **57600**
- Line ending: **Newline (`\n`)**

ESP32-S3 Mini variant (firmware v6.0)

Install ESP32 board support (one-time):

- File → Preferences → Additional Board Manager URLs
- Add: `https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json`
- Tools → Board → Board Manager → search for "ESP32" and install

Arduino IDE configuration:

- Board: **LOLIN S3 Mini** (or **ESP32S3 Dev Module**)
- USB CDC On Boot: **Enabled** (critical for Serial output!)
- USB Mode: **Hardware CDC and JTAG**
- Upload Mode: **UART0 / Hardware CDC**
- Flash Size: **4MB** or **8MB** (depending on the board)
- Library: install **FastLED** (3.5.0 or newer)

Upload:

1. Connect ESP32-S3 Mini to PC via USB — if upload fails: hold the **BOOT** button while connecting
2. Open the sketch `RGB_DPL-ESP32_S3_Mini_v6.0.ino`
3. Click **Upload**
4. After upload: press the **RESET** button on the ESP32-S3

Serial terminal after flashing:

- Baud rate: **9600**
- Line ending: **Newline (`\n`)**

ESP32-WROOM variant (RGB-DPL32, firmware v6.1)

For the older RGB-DPL hardware (up to HW REV 1.3) with a classic ESP32-WROOM.

Arduino IDE configuration:

- Board: **ESP32 Dev Module** (or the matching WROOM / D1-Mini ESP32 board)
- Flash Size: **4MB**, PSRAM: **off**
- Library: install **FastLED** (3.10.3 or newer)

Upload:

1. Connect the ESP32-WROOM to PC via USB
2. Open the sketch `RGB-DPL32.ino`
3. Click **Upload** (if the upload does not start: hold the **BOOT** button while connecting)

Serial terminal after flashing:

- Baud rate: **9600**
- Line ending: **Newline (`\n`)**

After flashing

- EEPROM settings (profiles, color schemes, personalities) are usually preserved
- If a firmware update changes the memory layout, profiles are reset to defaults automatically — note settings beforehand with `STATUS`

---

5. Command syntax

Commands are sent via the **serial interface** of the microcontroller. The RGB-DPL uses a **word-based command set** (no `/MODULE/PARAM/` format).

Properties:

- **Baud rate:** 57600 (Pro Mini) or 9600 (ESP32-S3/WROOM)
- **Line ending:** Newline (`\n`) — typical "Newline" or "NL" in your terminal
- **Case sensitivity:** none — commands are converted to uppercase internally

Format:

```
COMMAND [argument(s)]
```

If the board responds with a status text (e.g. `Brightness: 50`), the command was accepted. On errors you get an `ERROR:` message with a hint.

Examples:

Command	Effect
`BRIGHTNESS 50`	Master brightness 50%
`SCHEME CYBERPUNK`	Color scheme to Cyberpunk
`S4`	Quick command: color scheme 4 (Cyberpunk)
`PERSONALITY 1`	Personality to Happy
`P3`	Quick command: personality 3 (Excited)
`LDPLMODE 1`	LDPL to Rainbow

Command	Effect
`CBIMODE 6`	D-CBI to Matrix Rain
`CSLMODE 4`	CSL to Running Lights 1
`BARGRAPH SPLIT`	Bargraph two separate sides
`TOPBLOCKS 3`	Top Blocks Chaos mode (ESP32-S3/WROOM)
`SAVE 1`	Save current settings as profile 1
`LOAD 2`	Load profile 2
`STATUS`	Print current settings
`HELP`	Display the full command list

How to configure a profile — step by step

Example: a show setup for conventions with bright Cyberpunk look, then save as profile 1.

1. **Crank up brightness:**
2. **Pick the color scheme:**
3. **Personality to Excited:**
4. **Tune the animations:**
5. **Enable sequence mode** (auto-cycles between modes every 15 s):
6. **Save as profile 1:**
7. **Auto-load on next boot:**

> **Tip:** If you've lost track of your configuration, just load a known profile with `LOAD 1` (or whichever you saved). For total chaos: `DEFAULT 0` disables auto-load, then on next boot the firmware starts with built-in defaults.

---

6. LED strips A through G

The firmware addresses seven independent LED strips. The layout is identical on both controller variants — only the pin assignment differs.

Strip	LEDs	Pin (Pro Mini / ESP32-S3)	Function	Where physically?
A	8	D2 / IO4	Bottom White (6) + Red Large (2)	DPL board
B	28	D3 / IO5	VU meter	DPL board
C	24	D9 / IO6	Blue column (6) + bargraph (18)	DPL board
D	18	D8 / IO7	TopBlocks — yellow (9) + green (9)	DPL board
E	43	D7 / IO8	LDPL (Large DataPanel Logics)	**external**
F	23	D6 / IO9	D-CBI — matrix (20) + status (3)	**external**

Strip	LEDs	Pin (Pro Mini / ESP32-S3)	Function	Where physically?
G	36	D5 / IO13	CSL (Coin Slot Logics)	**external**

Total: 180 LEDs.

LED sections for the `COLOR` command

With the `COLOR <section> <R> <G> <B>` command you can paint individual sections with custom colors. The color scheme automatically switches to `CUSTOM` when you do this.

Section	Index	Default color (Classic)
Top Yellow Blocks	0	Yellow
Top Green Blocks	1	Green
Blue Column	2	Blue
Bargraph Low	3	Green
Bargraph Mid	4	Yellow
Bargraph High	5	Red
Bottom White	6	White
Red Large	7	Red
VU Meter	8	(palette-driven)
CBI Matrix	9	Red
CBI Status	10	(voltage-driven)
LDPL	11	Blue
CSL	12	Blue

Example: `COLOR 0 255 100 0` sets the Top Yellow Blocks to a bright orange.

---

7. Animation modes

LDPL (Strip E, 43 LEDs)

ESP32-S3 / WROOM (from firmware v6.1) — 16 modes:

`LDPLMODE`	Name	Description
0	Off	All LEDs off
1	Dot bounce	A single dot bounces back and forth
2	Fade trail	Bouncing dot with a fading trail
3	Rainbow dot	Bouncing dot cycling through rainbow colors
4	KITT / KnightRider	Clean scanner dot, bounces back and forth (no trail)
5	Cylon	Scanner with a soft trailing glow

LDPLMODE	Name	Description
6	Beat bars	Pulsing bars
7	Breathing	Gentle brightness pulsing in the color
8	Juggle	Several crossing dots
9	Sparkle	Random twinkle
10	Comet	Comet with a long tail
11	Alarm	Warning flash
12	Fire	Fire / flicker effect
13	Logic sweep	Logic-panel sweep
14	Rainbow bars	Rainbow bars across all 43 LEDs
15	Theater chase	Marching chase light

Set the base color of the single-color modes (e.g. KITT, Cylon) with `LDPLCOLOR <R> <G> <B>` (0-255, default red). `SAVE` stores mode **and** color in the profile. `LLMODE` is a legacy alias for `LDPLMODE`; the hotkeys `L0`–`L15` select the mode directly. Also controllable from the local web UI (chapter 14).

> **Pro Mini (v5.2):** keeps the previous 4 modes (`LDPLMODE 0-3` = Breathing, Rainbow, Off, Personality), no KITT/Cylon, no `LDPLCOLOR`, no web UI.

CBI / D-CBI (Strip F, 20 matrix LEDs + 3 status)

CBIMODE	Name	Description
0	Organic	Film-style random flickering in the scheme color (default)
1	ESB	Original 28-frame pattern sequence from Empire Strikes Back
2	Rainbow	Rainbow gradient across the matrix
3	Sparkle	Twinkling LEDs with hue variation and fade
4	Heart	Static heart pattern in red
5	Smiley	Static smiley face in yellow
6	Matrix Rain	Green "digital rain" cascade effect

> **Note:** With certain personalities/schemes the firmware overrides the CBI display: with **Happy** personality a smiley appears periodically, with **Pink** scheme a heart pattern.

CBI speed with `CBISPEED 1-10`.

CSL (Strip G, 36 LEDs — external module)

CSLMODE	Name	Description
0	Off	All LEDs off
1	Default	Running lights with constant spacing (default)
2	Static	All LEDs on in the scheme color

CSLMODE	Name	Description
3	Glow	Pulsing brightness glow
4	Running Lights 1	Running lights with 3-LED spacing
5	Running Lights 2	Two running lights, one from each side
6	Running Lights 3	3-LED block running lights
7	Running Lights 4	Sine-modulated pixel sweep with trail

CSL speed: `CSLSPEED 1-10`. **CSL brightness (relative):** `CSLBRIGHTNESS 1-100`.

TopBlocks (Strip D, 18 LEDs)

ESP32-S3 Mini (v6.0) — 5 modes:

TOPBLOCKS	Name	Description
0	Classic	Random paired block activation (film-accurate)
1	Horizontal	Full row illumination
2	Paired	Two adjacent rows illuminate together
3	Chaos	Random individual pixel per column
4	Single Pixel	One pixel per color section

Pro Mini (v5.2) — 2 modes:

TOPBLOCKS	Name	Description
`RANDOM`	Random	Organic random activation using bit patterns
`CLASSIC`	Classic	One random pixel per color section

Bargraph (within Strip C, 18 LEDs)

BARGRAPH	Description
`CLASSIC`	Both sides show the same random level (mirrored)
`SPLIT`	Left and right sides animate independently

VU meter (Strip B, 28 LEDs)

The VU meter is **automatically** driven by the active color scheme + personality — no separate mode command. Examples:

- **Pink scheme:** pulsating pink palette
- **Happy personality:** rainbow gradient
- **Sleepy personality:** pulsing dim blue
- **Otherwise:** classic VU bargraph with `beatsin8` pulses

Quick commands (single-character hotkeys)

Command	Function
`L0`-`L3`	Set LDPL mode
`C0`-`C6`	Set CBI mode
`P0`-`P4`	Set personality
`S0`-`S7`	Set color scheme

---

8. Color schemes

The firmware ships with 8 predefined color schemes that re-color all LED sections at once:

Index	Name	Description
0	CLASSIC	Film-accurate colors (Yellow, Green, Blue, Red, White)
1	BLUE	Cool blue tones across all panels
2	PINK	Pink/Magenta theme (triggers heart pattern on CBI)
3	GREEN	Forest green variations
4	CYBERPUNK	Magenta/Cyan/Purple neon look
5	FOREST	Earth tones with orange and dark green
6	SUNSET	Warm orange/red/violet gradients
7	CUSTOM	User-defined via the `COLOR` command

Set with:

- `SCHEME 4` (numeric)
- `SCHEME CYBERPUNK` (name)
- `S4` (quick command)

Tip: With `SCHEME CUSTOM` you can color each LED section individually — see the `COLOR` command in *Section 6*.

---

9. Personality modes

Personalities modify global animation speed, CBI timing, hue shift and the "organic" behavior:

Index	Name	Speed	CBI speed	Hue shift	Behavior
0	NORMAL	1.0×	1.0×	0	Standard R2-D2 behavior
1	HAPPY	1.5×	1.2×	+30	Faster, warmer, rainbow VU, periodic smiley on CBI
2	GRUMPY	0.5×	0.8×	-30	Slower, cooler tones, dim pulsing

Index	Name	Speed	CBI speed	Hue shift	Behavior
3	EXCITED	2.0×	1.5×	0	Very fast, energetic rainbow LDPL
4	SLEEPY	0.3×	0.5×	-60	Very slow, dim blue breathing

Set with:

- `PERSONALITY 1`
- `P1` (quick command)

---

10. Profiles (SAVE / LOAD / DEFAULT)

The firmware has **5 profile slots** in EEPROM with checksum protection. Each profile stores:

- Profile name (8 chars Pro Mini / 16 chars ESP32-S3/WROOM)
- Brightness, animation speed, CBI speed
- Color scheme + personality
- LDPL mode, CBI mode, CSL mode
- Bargraph style, TopBlocks mode
- Sequence and voltage monitor status
- CSL speed + CSL brightness
- 12 custom colors (for scheme `CUSTOM`)

Commands

Command	Effect
`SAVE 1` to `SAVE 5`	Save current settings to profile 1-5
`LOAD 1` to `LOAD 5`	Load profile 1-5
`DEFAULT 0`	No auto-load on boot (built-in defaults)
`DEFAULT 1` to `DEFAULT 5`	Auto-load this profile on boot

Default values

Parameter	Default
Brightness	25%
DPL Speed	5
CBI Speed	3 (ESP32-S3/WROOM) / 5 (Pro Mini)
Color Scheme	CLASSIC (0)
Personality	NORMAL (0)
LDPL Mode	0 (Breathing)
CBI Mode	0 (Organic)
CSL Mode	1 (Default)

Parameter	Default
Bargraph	Classic (mirrored)
TopBlocks	Classic (0) on ESP32-S3/WROOM / RANDOM on Pro Mini
Sequence Mode	OFF
Voltage Monitor	OFF
CSL Speed	5
CSL Brightness	25%

---

11. Configuring the voltage monitor

The RGB-DPL can measure an external battery voltage and indicate the charge state on the three status LEDs of the D-CBI.

> **Important:** The `Voltage IN` input is **separate** from the `5V IN` supply — the board runs on 5V, the Voltage IN is exclusively a high-impedance measurement input.

> **Off by default from v6.1:** On the ESP32-S3 / WROOM variants the measurement input shares the analog path (ADC2) with Wi-Fi. Because the local web UI (chapter 14) uses Wi-Fi, the voltage monitor is **off by default**. Enable it with `VM 1`; reliable values are only available while the access point is **off** (after the timeout or `AP OFF`).

Activate

```
VOLTAGE ON              # enable CBI status LED display
VOLTAGE OFF             # disable
VM 1                    # global VCC monitor on (alternative form)
VM 0                    # off
```

Set voltage thresholds

```
VL <RED> <YELLOW> <GREEN> <CHARGE>
```

Examples:

```
VL 11.5 12.0 12.5 13.0       # 12V lead-acid (default)
VL 10.8 11.4 12.0 12.6       # 3S LiPo
VL 23.0 22.5 22.0 25.0       # 24V system (with adapted divider)
```

Threshold	Meaning
RED	Below: red status LED on, warning
YELLOW	Between RED and YELLOW: yellow
GREEN	Between YELLOW and GREEN: green (normal)
CHARGE	Above: charging indicator (charger detected)

Voltage divider values (hardware adaptation)

```
RR <R1> <R2>
```

Default: R1 = 100 kΩ, R2 = 10 kΩ (for 0–~30 V battery measurement).

Examples:

```
RR 100000 10000          # default
RR 220000 33000          # higher voltage range, higher resolution at low V (legacy v5.8 ESP32)
```

Important: The `RR` command only changes the software scaling (how the firmware interprets the ADC value), **not** the hardware. If you replace the SMD resistors, set `RR` to the actual values, otherwise the displayed voltage will be wrong.

Display

- **On D-CBI** (3 dedicated WS2812B LEDs): green/yellow/red depending on voltage
- **Charging animation** is started automatically when the voltage rises above `CHARGE`

---

12. Door sensors (LeftDoor / RightDoor)

The RGB-DPL has two door switch inputs on dedicated 2-pin headers. Both have internal pull-ups — a switch only needs to **pull the pin to GND**.

Input	Header	Pin (Pro Mini / ESP32-S3)	Pin LOW (to GND)	Pin HIGH (not connected)
Left Door	2-pin left	A2 (16) / IO10	DPL strips A, B, C, D run	DPL strips A-D all off
Right Door	2-pin right	A1 (15) / IO11	CBI strip F runs	CBI all off

Note: Strip E (LDPL) is **always active**, regardless of door switch state.

How to activate the switches

Option 1 — Reed switch with magnet (recommended)

- NO (Normally Open) reed contact — closes when a magnet is near
- Wiring: one pin to the header pin (door signal), the other to GND
- Position the magnet so that it's **next to the reed when the door is open** → reed closes → pin = LOW → DPL runs

Option 2 — Mechanical micro switch

- NO switch between header pin and GND
- Door open → pin LOW → DPL runs

Option 3 — Hall effect sensor

- Digital hall sensor (e.g. A3144 with open-drain output)
- VCC = 5V, GND = GND, output to door header

Option 4 — External controller / body master

- Other microcontroller pulls door signal to GND or leaves floating
- Both sides need the same GND

Option 5 — Wire jumper (LEDs always on)

- Direct between the two pins of each header — pin permanently LOW → all LEDs always active
- Useful for show setup without movable doors or for testing

Behavior table

Left Door	Right Door	Active LEDs
Open (LOW)	Open (LOW)	All strips A-G
Open (LOW)	Closed (HIGH)	A, B, C, D + LDPL
Closed (HIGH)	Open (LOW)	F (CBI) + LDPL
Closed (HIGH)	Closed (HIGH)	LDPL only

---

13. System commands

Command	Effect
`HELP`	Show full command list (with ANSI colors on ESP32-S3/WROOM)
`STATUS`	Print current settings (profile, brightness, modes, etc.)
`INFO`	System info (version, free heap, boot count) — **ESP32-S3/WROOM only**
`CONFIG`	Enter configuration mode (LDPL blinks blue as indicator)
`EXIT`	Exit configuration mode
`SAVE 1-5`	Save settings to profile
`LOAD 1-5`	Load profile
`DEFAULT 0-5`	Set boot profile (0 = off)

> Settings are kept in the current profile buffer for every command — **only** `SAVE` writes them to EEPROM. If you want to test settings without keeping them, just don't run `SAVE` and they're lost on reboot.

---

14. Local web UI & access point

From firmware **6.1.0** (the ESP32-S3 variants and the WROOM variant **RGB-DPL32**) the board ships with a **local web interface** that shows all key settings in a browser — no serial console needed. The web UI mirrors the serial commands: whatever you click runs exactly the same command internally.

> The Pro Mini / Nano variant (v5.x) has **no** web UI — the ATmega328P has no WiFi module.

Connecting

1. After power-on the board opens its own Wi-Fi access point (SoftAP). The network name (SSID) starts with `RGB-DPL` (the three variants differ in the name).
2. Join that Wi-Fi from your phone or laptop.
3. Open **http://192.168.4.1** in the browser.

The AP is **open at first** (no password). Set your own WPA2 password with `APPW <password>` (8–63 characters); `APPW CLEAR` makes it open again. Changing the password restarts the AP and disconnects the current client. The password is never shown in the status output.

Automatic shutdown (save power/RF)

If nobody connects within **2 minutes** (default) the access point switches **off** automatically. While a device is connected it stays on; after it disconnects the timer restarts.

```
APTIMEOUT <seconds>     # shutdown time (0-86400); 0 = never shut down
AP ON                   # turn the access point on
AP OFF                  # turn the access point off
```

A reboot briefly brings the AP back up anyway.

What the web UI controls

LDPL mode (incl. KITT/Cylon) + a **color picker** for the LDPL base color, CBI mode, TopBlocks, color scheme (incl. custom section colors), personality, brightness and speeds, sequence mode, bargraph, the 5 profiles + boot default, voltage thresholds/divider and — if populated — CSL. The door-gating display shows the physical switch state.

Web UI/Wi-Fi and voltage measurement

The voltage measurement input sits on an ADC2 pin, and **ADC2 cannot be used while Wi-Fi is active**. That is why voltage monitoring is **off by default** (see chapter 11). Once enabled with `VM 1` it only returns values when the access point is **off** (e.g. after the 2-minute timeout or `AP OFF`) — while the AP is active no (wrong) values are shown.

---

15. Troubleshooting

Command not accepted (no "OK")

1. **Check baud rate** — Pro Mini: 57600, ESP32-S3/WROOM: 9600. Wrong baud rate → garbage in terminal.
2. **Check line ending** — Newline (`\n`), not "No line ending" or CR-only.
3. **For ESP32-S3:** "USB CDC On Boot" must be set to "Enabled" in the Arduino IDE.
4. **Command spelled correctly?** Typo? Commands are case-insensitive, but arguments must match.
5. **Command length** — max. 24 characters (Pro Mini) or 64 characters (ESP32-S3/WROOM).

LEDs don't light up

- **Door switches HIGH?** → Certain strips stay dark. Set jumper or connect reed switches.
- **Brightness 0?** → set `BRIGHTNESS 50`.
- **Animation 0 (off)?** → `LDPLMODE 0` leaves LDPL on (Breathing), `CBIMODE 0` is Organic. Check with `STATUS`.
- **Power supply?** Very bright setups (all 180 LEDs full) need 4A at 5V.
- **Data line broken?** First LED of the external strip has to be on the pin (DCBI=D6/IO9, LDPL=D7/IO8, CSL=D5/IO13), then daisy-chain.

Wrong colors

WS2811/WS2812B come in several color orderings (RGB, GRB, BGR). The firmware is compiled for **GRB**. Strips with a different ordering show swapped colors — solution: buy matching strips or change the firmware source (`FastLED.addLeds<WS2811, ..., GRB>(...)` to `RGB` etc.).

Voltage display shows wrong value

1. **Check voltage divider values** — default 100k/10k. If you replaced the SMD resistors, set `RR <R1> <R2>` to the actual values.
2. **Check threshold values** — `VL RED YELLOW GREEN CHARGE`. Order must be ascending.
3. **ADC voltage** — the ADC pin must see at most 5V (Pro Mini) or 3.3V (ESP32-S3/WROOM). For high battery voltages: use a suitable divider.

EEPROM settings lost after update

The firmware uses a checksum (`0xDEAD` magic) for integrity validation. If an update changes the profile layout, the profile is detected as invalid and built-in defaults take over.

Solution: Before updating, note key values with `STATUS`, after updating set them again and `SAVE 1` back into the profile.

Watchdog reset (ESP32-S3/WROOM)

If the firmware hangs in an animation, the watchdog resets after 10 seconds. Looks like a boot loop. Should not normally happen — if it does, that's a firmware bug. Please report via GitHub issues.

"ESP-ROM:esp32s3" / no serial output (ESP32-S3)

In the Arduino IDE, "USB CDC On Boot" must be set to **Enabled**. Otherwise the ESP32-S3 only outputs the ROM boot text.

---

16. FAQ

Which variant do I have?

Look at the back of the RGB-DPL: a plug-in microcontroller module. **Pro Mini** = ATmega328P module (oblong, no USB-C). **ESP32-S3 Mini** = ESP32 module with USB-C connector. You can run both variants on the same board — only the firmware has to match.

What happens without D-CBI / LDPL / CSL?

Nothing bad — the firmware drives the pins anyway, the connected module (or none) decides what lights up. Commands like `CBIMODE 3` are still valid, you just don't see the animation.

Can I use D-CBI and CBI Plus (analog MAX7219 variant) at the same time?

No — the CBI Plus module (MAX7219) is not supported by the RGB-DPL firmware. The RGB-DPL uses purely WS2811/WS2812 LEDs, so only the **D-CBI**.

What's the difference between `LDPLMODE` and `LLMODE`?

`LLMODE` is a legacy alias for `LDPLMODE` from older firmware versions. Both commands do the same thing — use whichever you're used to.

What is "Sequence mode"?

With `SEQUENCE ON` you activate auto-cycling: every 15 seconds the firmware automatically switches between different animation modes. Handy for conventions / shows where you don't want to switch manually all the time.

Can I run both DPL versions (Pro Mini + ESP32-S3) in parallel?

Yes, if you actually have two boards. Each board is autonomous and has its own serial console. Not needed if you only have one R2 — one RGB-DPL is enough.

How much current do how many LEDs draw?

Worst-case estimate at full brightness, all 180 LEDs on white: ~10-11A at 5V. With default brightness (25%) and mixed colors: ~1-2A. The firmware has a FastLED power limiter set to 5V/4A (ESP32-S3/WROOM) or 5V/2A (Pro Mini), which automatically dims if needed to stay below that.

Is there a Web UI?

No. The RGB-DPL is configured exclusively via the serial interface.

How do I integrate the RGB-DPL with MarcDuino / BetterDuino?

ESP32-S3 variant: I2C slave address `0x20` is active (SDA/SCL = default ESP32-S3 pins). Currently the I2C routing is a stub — the interface is prepared for future MarcDuino integration but commands aren't routed yet.

Pro Mini variant: I2C was disabled in v5.2 (not enough flash). In earlier versions address `0x20` was active.

For serial integration with MarcDuino: the default baudrate (57600 / 9600) has to be changed to 9600 in the source and recompiled, since MarcDuino speaks 9600.

Where do I find current firmware HEX files?

In the GitHub repo: https://github.com/PrintedDroid/RGB-DPL-Firmware

Where do I get help?

- **Product page:** https://www.printed-droid.com
- **Facebook group:** https://www.facebook.com/groups/printeddroid/
- **GitHub repo:** https://github.com/PrintedDroid/RGB-DPL-Firmware

---

17. Cheat sheet (quick reference)

A one-page summary to print and keep next to your soldering bench / PC. The ESP32-S3 variant has the full command set; the Pro Mini variant is a subset (see Section 16.2).

General

```
Command format: COMMAND [argument(s)]    (case-insensitive)
Line ending:    Newline (\n)
LED strips:     7 (A-G), 180 LEDs total
Profiles:       5 slots (1-5)
```

17.1 ESP32-S3 / WROOM (v6.1) — Full command reference

Baud rate: 9600

System commands

Command	Effect
`HELP`	Full command list (with ANSI colors)

Command	Effect
`STATUS`	Print current settings
`INFO`	System info (version, free heap, boot count)
`CONFIG` / `EXIT`	Config mode on/off
`SAVE 1-5`	Save settings to profile
`LOAD 1-5`	Load profile
`DEFAULT 0-5`	Set boot profile (0 = off)

Global settings

Command	Range
`BRIGHTNESS <1-100>`	master in %
`SPEED <1-10>`	DPL animation
`CBISPEED <1-10>`	CBI animation
`CSLSPEED <1-10>`	CSL animation
`CSLBRIGHTNESS <1-100>`	CSL relative
`SCHEME <name|0-7>`	color scheme
`PERSONALITY <0-4>`	NORMAL/HAPPY/GRUMPY/EXCITED/SLEEPY
`COLOR <0-12> <R> <G> <B>`	custom color (switches to CUSTOM)
`SEQUENCE ON|OFF`	auto-cycling
`VOLTAGE ON|OFF`	CBI voltage display

Animation modes

```
LDPLMODE 0-15     0:Off 1:DotBounce 2:FadeTrail 3:RainbowDot 4:KITT 5:Cylon 6:BeatBars
                  7:Breathe 8:Juggle 9:Sparkle 10:Comet 11:Alarm 12:Fire 13:LogicSweep
                  14:RainbowBars 15:TheaterChase   (Pro Mini: 0-3 only)
LLMODE 0-15       alias for LDPLMODE
LDPLCOLOR R G B   base color for the LDPL modes (0-255, default red); SAVE persists
CBIMODE  0-6      0:Organic 1:ESB 2:Rainbow 3:Sparkle 4:Heart 5:Smiley 6:Matrix
CSLMODE  0-7      0:Off 1:Default 2:Static 3:Glow 4-7:Running
BARGRAPH SPLIT|CLASSIC
TOPBLOCKS 0-4     0:Classic 1:Horizontal 2:Paired 3:Chaos 4:Single
```

Voltage monitor

```
VM 0|1                              # monitor on/off
VL <RED YELLOW GREEN CHARGE>        # thresholds in V (default: 11.5/12.0/12.5/13.0)
RR <R1 R2>                          # voltage divider in ohms (default: 100000/10000)
```

Quick commands (single-character hotkeys)

```
P0-P4    personality
S0-S7    color scheme
L0-L15   LDPL mode   (Pro Mini: L0-L3)
C0-C6    CBI mode
```

Color scheme indices

```
0 CLASSIC   1 BLUE      2 PINK      3 GREEN
4 CYBERPUNK 5 FOREST    6 SUNSET    7 CUSTOM
```

COLOR sections

```
0 TopYellow  1 TopGreen   2 BlueColumn  3 BargraphLow  4 BargraphMid
5 BargraphHi 6 BottomWh   7 RedLarge    8 VuMeter      9 CbiMatrix
10 CbiStatus 11 LDPL      12 CSL
```

17.2 Pro Mini (v5.2) — Differences from the ESP32 variant (S3/WROOM)

Baud rate: 57600 (instead of 9600)

Otherwise the **same commands** as ESP32-S3, **except for:**

What's not available

Command / feature	Status	Note
`INFO`	■ not implemented	use `STATUS`
I2C slave (0x20)	■ disabled	removed in v5.2 to save flash
Watchdog	■ not active	present in v5.1, removed in v5.2

What's different

Command	Pro Mini	ESP32-S3
`TOPBLOCKS`	`RANDOM` or `CLASSIC` only (2 modes)	`0`-`4` (5 modes)
`LDPLMODE` / `L` hotkeys	`0-3` (4 modes, no KITT/Cylon, no `LDPLCOLOR`)	`0-15` (16 modes incl. KITT/Cylon) + `LDPLCOLOR`
Local web UI / access point	not available	available from v6.1
Profile name length	8 characters	16 characters
Boot sequence	only 3× white flash	rainbow wave + 3× white flash

All other commands (BRIGHTNESS, SPEED, SCHEME, PERSONALITY, COLOR, SEQUENCE, VOLTAGE, CBIMODE, CSLMODE, CSLSPEED, CSLBRIGHTNESS, BARGRAPH, VM, VL, RR, P0-P4, S0-S7, C0-C6, SAVE, LOAD, DEFAULT, HELP, STATUS, CONFIG, EXIT) behave the same as on the ESP32-S3/WROOM variant. `LDPLMODE`/`LLMODE`/`L` hotkeys and `LDPLCOLOR` differ (see the table above).

Example setups (for both variants)

```
# Convention / show – colorful + fast
BRIGHTNESS 80 / SCHEME CYBERPUNK / PERSONALITY 3 / SEQUENCE ON
LDPLMODE 1 / CBIMODE 2 / CSLMODE 7
SAVE 1 / DEFAULT 1

# Stealth / dimmed
BRIGHTNESS 15 / SCHEME BLUE / PERSONALITY 4 / SEQUENCE OFF
LDPLMODE 0 / CBIMODE 0 / CSLMODE 3
SAVE 2

# Classic R2-D2
BRIGHTNESS 30 / SCHEME CLASSIC / PERSONALITY 0
LDPLMODE 0 / CBIMODE 0 / CSLMODE 1 / BARGRAPH CLASSIC
SAVE 3

# Convention TopBlocks (variant-dependent)
ESP32-S3:   TOPBLOCKS 3        # Chaos mode
Pro Mini:   TOPBLOCKS RANDOM   # equivalent mode
```

---

18. Appendix: Legacy firmware v5.1 / v5.8

> **Getting current versions:** New firmware for the **ESP32-S3 / ESP32-WROOM** variants (v6.1, incl. Web UI / Access Point) is released only via the Printed-Droid **web flasher**. The **Nano / Pro Mini** firmware is **final at v5.2** and lives only in the GitHub repo `PrintedDroid/RGB-DPL-Firmware` — there are no new Nano builds and it is not on the web flasher. The legacy versions described here are documented for existing devices only.

For users running an older firmware version: here's what differs from the current versions.

v5.1 (Pro Mini) — predecessor of v5.2

What's missing:

- **CSL module** entirely — pin D5 is not driven. An external CSL board stays dark with v5.1. For CSL support, update to v5.2.
- The commands `CSLMODE`, `CSLSPEED`, `CSLBRIGHTNESS` are not implemented.

What's different:

- **TopBlocks:** uses the strings `RANDOM`/`CLASSIC` (same as v5.2). 2 modes.
- **I2C slave active** on address `0x20` (removed in v5.2 to save flash).
- Low-voltage warning: all 3 CBI status LEDs blink red (300 ms) when below the `RED` threshold.

Migration v5.1 → v5.2: profiles are usually preserved because the layout is compatible. CSL just starts running.

v5.8 (ESP32-S3) — predecessor of v6.0

What's missing:

- **CSL module** entirely.
- The commands `CSLMODE`, `CSLSPEED`, `CSLBRIGHTNESS` are not implemented.
- The section index field only goes up to 11 (no 12 for CSL).

What's different:

- **Voltage divider default:** R1 = 220 kΩ, R2 = 33 kΩ (vs. 100k/10k on v6.0). If you have v5.8 hardware and update to v6.0, set `RR 220000 33000` to match the old values, or replace the SMDs.
- **System voltage default:** 24V (vs. 12V on v6.0). Use `VL` to match your system.
- **Baud rate:** 57600 (vs. 9600 on v6.0). Watch out when updating, adjust your terminal.
- **TopBlocks:** already 5 modes (same as v6.0).

Migration v5.8 → v6.0: because of the changed baud rate and VCC defaults, note settings beforehand with `STATUS`, after flashing change the terminal baud rate to 9600, then reconfigure `VL` and `RR` if needed, finally `SAVE` the profile again.